Lista teórica - 1

Atividade prática:

- Dataset: escolher um relacional no Kaggle (diferente do Olist);
- Usar MySQL ou PostgreSQL;
- Executar consultas e registrar em lista1.sql e arquivos com resultados;
- Cada Consulta deve ter comentário com objetivo.

Requisitos mínimos:

- No mínimo 6 consultas;
- No mínimo 3 consultas com JOIN;
- No mínimo 2 consultas com GROUP BY;
- No mínimo 1 consulta com HAVING;
- No mínimo 1 consulta com ORDER BAY e LIMIT.

1 - Listar filmes e suas avaliações. Objetivo: relacionar os filmes com as notas dadas pelos usuários. (Consulta com JOIN)

```
SELECT m.title, r.userId, r.rating

FROM movies m

JOIN ratings r ON m.movieId = r.movieId

LIMIT 20;
```

```
mysql> SELECT m.title, r.userId, r.rating
    -> FROM movies m
    -> JOIN ratings r ON m.movieId = r.movieId
    -> LIMIT 20;
+----------------------------------------+--------+--------+
| title                                  | userId | rating |
+----------------------------------------+--------+--------+
| Toy Story (1995)                       |      1 |      4 |
| Grumpier Old Men (1995)                |      1 |      4 |
| Heat (1995)                            |      1 |      4 |
| Seven (a.k.a. Se7en) (1995)            |      1 |      5 |
| Usual Suspects, The (1995)             |      1 |      5 |
| From Dusk Till Dawn (1996)             |      1 |      3 |
| Bottle Rocket (1996)                   |      1 |      5 |
| Braveheart (1995)                      |      1 |      4 |
| Rob Roy (1995)                         |      1 |      5 |
| Canadian Bacon (1995)                  |      1 |      5 |
| Desperado (1995)                       |      1 |      5 |
| Billy Madison (1995)                   |      1 |      5 |
| Clerks (1994)                          |      1 |      3 |
| Dumb & Dumber (Dumb and Dumber) (1994) |      1 |      5 |
| Ed Wood (1994)                         |      1 |      4 |
| Star Wars: Episode IV - A New Hope (1977) |      1 |      5 |
| Pulp Fiction (1994)                    |      1 |      3 |
| Stargate (1994)                        |      1 |      3 |
| Tommy Boy (1995)                       |      1 |      5 |
| Clear and Present Danger (1994)        |      1 |      4 |
+----------------------------------------+--------+--------+
20 rows in set (0.00 sec)
```

2 - Média de avaliação de cada filme. Objetivo: calcular a média das notas recebidas por cada filme. (Consulta com JOIN + GROUP BY)

```
SELECT m.title, AVG(r.rating) AS media_avaliacao

FROM movies m

JOIN ratings r ON m.movieId = r.movieId GROUP BY

m.title

ORDER BY media_avaliacao

DESC LIMIT 10;
```

```
mysql> SELECT m.title, AVG(r.rating) AS media_avaliacao
    -> FROM movies m
    -> JOIN ratings r ON m.movieId = r.movieId
    -> GROUP BY m.title
    -> ORDER BY media_avaliacao DESC
    -> LIMIT 10;
+-------------------------------------------------------+-----------------+
| title                                                 | media_avaliacao |
+-------------------------------------------------------+-----------------+
| Bill Hicks: Revelations (1993)                        |               5 |
| The Jinx: The Life and Deaths of Robert Durst (2015) |               5 |
| Assignment, The (1997)                                |               5 |
| Black Mirror                                          |               5 |
| Mephisto (1981)                                       |               5 |
| Lesson Faust (1994)                                   |               5 |
| Galaxy of Terror (Quest) (1981)                       |               5 |
| Alien Contamination (1980)                            |               5 |
| I'm the One That I Want (2000)                        |               5 |
| Dylan Moran: Monster (2004)                           |               5 |
+-------------------------------------------------------+-----------------+
10 rows in set (0.39 sec)
```

3 - Quantidade de avaliações por filme. Objetivo: descobrir quais filmes possuem mais avaliações. (GROUP BY + ORDER BY + LIMIT)

```
SELECT m.title, COUNT(r.rating) AS total_avaliacoes

FROM movies m

JOIN ratings r ON m.movieId = r.movieId

GROUP BY m.title

ORDER BY total_avaliacoes

DESC LIMIT 10;
```

```
mysql> SELECT m.title, COUNT(r.rating) AS total_avaliacoes
    -> FROM movies m
    -> JOIN ratings r ON m.movieId = r.movieId
    -> GROUP BY m.title
    -> ORDER BY total_avaliacoes DESC
    -> LIMIT 10;
+-----------------------------------------+------------------+
| title                                   | total_avaliacoes |
+-----------------------------------------+------------------+
| Forrest Gump (1994)                     |              329 |
| Shawshank Redemption, The (1994)        |              317 |
| Pulp Fiction (1994)                     |              307 |
| Silence of the Lambs, The (1991)        |              279 |
| Matrix, The (1999)                      |              278 |
| Star Wars: Episode IV - A New Hope (1977) |              251 |
| Jurassic Park (1993)                    |              238 |
| Braveheart (1995)                       |              237 |
| Terminator 2: Judgment Day (1991)       |              224 |
| Schindler's List (1993)                 |              220 |
+-----------------------------------------+------------------+
10 rows in set (0.37 sec)
```

4 - Filmes com média de avaliação maior que 4.5. Objetivo: identificar filmes muito bem avaliados. (GROUP BY + HAVING)

SELECT m.title, AVG(r.rating) AS media_avaliacao

FROM movies m

JOIN ratings r ON m.movieId = r.movieId

GROUP BY m.title

HAVING AVG(r.rating) > 4.5

ORDER BY media_avaliacao DESC;

```
mysql> SELECT m.title, AVG(r.rating) AS media_avaliacao
    -> FROM movies m
    -> JOIN ratings r ON m.movieId = r.movieId
    -> GROUP BY m.title
    -> HAVING AVG(r.rating) > 4.5
    -> ORDER BY media_avaliacao DESC;
+-----------------------------------------------------------------------------------------+-------------------+
| title                                                                                   | media_avaliacao   |
+-----------------------------------------------------------------------------------------+-------------------+
| The Jinx: The Life and Deaths of Robert Durst (2015)                                    |                 5 |
| Galaxy of Terror (Quest) (1981)                                                         |                 5 |
| Alien Contamination (1980)                                                              |                 5 |
| I'm the One That I Want (2000)                                                          |                 5 |
| Lesson Faust (1994)                                                                     |                 5 |
| Assignment, The (1997)                                                                  |                 5 |
| Mephisto (1981)                                                                         |                 5 |
| Black Mirror                                                                            |                 5 |
| Dylan Moran: Monster (2004)                                                             |                 5 |
| Bill Hicks: Revelations (1993)                                                          |                 5 |
| My Sassy Girl (Yeopgijeogin geunyeo) (2001)                                             |                 5 |
| Strictly Sexual (2008)                                                                  |                 5 |
| Story of Women (Affaire de femmes, Une) (1988)                                          |                 5 |
| Passenger, The (Professione: reporter) (1975)                                           |                 5 |
| Marriage of Maria Braun, The (Ehe der Maria Braun, Die) (1979)                          |                 5 |
```

```
| Whiplash (2013)                                                                         |              4.75 |
| Imposter, The (2012)                                                                    |              4.75 |
| Victory (a.k.a. Escape to Victory) (1981)                                               |              4.75 |
| Best of Youth, The (La meglio giovent◆) (2003)                                          |              4.75 |
| Wild Parrots of Telegraph Hill, The (2003)                                              |              4.75 |
| Memories of Murder (Salinui chueok) (2003)                                              |               4.7 |
| Chorus Line, A (1985)                                                                   | 4.666666666666667 |
| Children of the Corn IV: The Gathering (1996)                                           | 4.666666666666667 |
| Swept Away (Travolti da un insolito destino nell'azzurro mare d'Agosto) (1975)          | 4.666666666666667 |
| Long Goodbye, The (1973)                                                                | 4.666666666666667 |
| Dog Soldiers (2002)                                                                     | 4.666666666666667 |
| Son of Rambow (2007)                                                                    | 4.666666666666667 |
| Autumn Sonata (H◆stsonaten) (1978)                                                      |             4.625 |
| No Direction Home: Bob Dylan (2005)                                                     |             4.625 |
| Tekkonkinkreet (Tekkon kinkur◆to) (2006)                                                |             4.625 |
| Sophie Scholl: The Final Days (Sophie Scholl - Die letzten Tage) (2005)                 |             4.625 |
| Yi Yi (2000)                                                                            |               4.6 |
| Secrets & Lies (1996)                                                                   | 4.590909090909091 |
| Day of the Doctor, The (2013)                                                           | 4.571428571428571 |
| Guess Who's Coming to Dinner (1967)                                                     | 4.545454545454546 |
| Paths of Glory (1957)                                                                   | 4.541666666666667 |
+-----------------------------------------------------------------------------------------+-------------------+
344 rows in set (0.38 sec)
```

5 - Filmes e suas tags associadas. Objetivo: visualizar quais tags os usuários associaram aos filmes. (JOIN entre movies e tags)

SELECT m.title, t.tag

FROM movies m

JOIN tags t ON m.movieId =

t.movieId ORDER BY m.title

LIMIT 20;

```
mysql> SELECT m.title, t.tag
    -> FROM movies m
    -> JOIN tags t ON m.movieId = t.movieId
    -> ORDER BY m.title
    -> LIMIT 20;
+-------------------------------------------------------+---------------------+
| title                                                 | tag                 |
+-------------------------------------------------------+---------------------+
| ...And Justice for All (1979)                         | lawyers             |
| (500) Days of Summer (2009)                           | quirky              |
| (500) Days of Summer (2009)                           | artistic            |
| (500) Days of Summer (2009)                           | intelligent         |
| (500) Days of Summer (2009)                           | Zooey Deschanel     |
| (500) Days of Summer (2009)                           | romance             |
| (500) Days of Summer (2009)                           | humorous            |
| (500) Days of Summer (2009)                           | Funny               |
| (500) Days of Summer (2009)                           | inspiring           |
| 10 Cloverfield Lane (2016)                            | suspense            |
| 10 Cloverfield Lane (2016)                            | creepy              |
| 10 Things I Hate About You (1999)                     | Shakespeare sort of |
| 101 Dalmatians (1996)                                 | dogs                |
| 101 Dalmatians (1996)                                 | remake              |
| 101 Dalmatians (One Hundred and One Dalmatians) (1961) | Disney              |
| 11'09"01 - September 11 (2002)                        | terrorism           |
| 12 Angry Men (1957)                                   | confrontational     |
| 12 Angry Men (1957)                                   | court               |
| 12 Angry Men (1957)                                   | earnest             |
| 12 Angry Men (1957)                                   | good dialogue       |
+-------------------------------------------------------+---------------------+
20 rows in set (0.01 sec)
```

6 - Usuários que mais avaliaram filmes. Objetivo: identificar quais usuários fizeram mais avaliações na base. (GROUP BY + ORDER BY + LIMIT)

```
SELECT userId, COUNT(rating) AS total_avaliacoes

FROM ratings


GROUP BY userId

ORDER BY total_avaliacoes DESC

LIMIT 10;
```

```
mysql> SELECT userId, COUNT(rating) AS total_avaliacoes
    -> FROM ratings
    -> GROUP BY userId
    -> ORDER BY total_avaliacoes DESC
    -> LIMIT 10;
+--------+------------------+
| userId | total_avaliacoes |
+--------+------------------+
|    414 |             2698 |
|    599 |             2478 |
|    474 |             2108 |
|    448 |             1864 |
|    274 |             1346 |
|    610 |             1302 |
|     68 |             1260 |
|    380 |             1218 |
|    606 |             1115 |
|    288 |             1055 |
+--------+------------------+
10 rows in set (0.12 sec)
```